OPQAN: Orthogonal Polynomial-Based Quantum Attention Network for Plant Leaf Disease Prediction

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***Abstract*—Diseases of crops are a big threat to food security globally, and they lead to a lot of losses in agriculture worldwide. Deep learning methods are already surfacing non-manual disease detection, but due to high demands for resources and unproven across different areas of application, these systems are still slow to gain acceptance. In this paper, a hybrid classical - quantum framework based Orthogonal Polynomial Transformation (OPT) and Quantum Attention Network (QAN) for the diagnosis plant disease. It combine low level classical features are automatically extracted from plant image based on Statistical, Texture and Convolutional Neural Network(CNN) methods. The OPQAN method instinctively extracts quantum features based on quantum parallelism and entanglements to retrieve relevant the subtle patterns of plant diseases. The proposed method tested on the PlantDoc dataset, and reaches 91.3 % accuracy in classifications, which is better than the baselines of the classical CNN (85.2 %) and MobileNetV2 (87.6 %). Our method decreases the total number of parameters by 14 %, while at the same time, generalization is enhanced. The proposed research confirms that the classical feature extraction method support of quantum computer vision in precision agriculture.**

***Index Terms*—Quantum Machine Learning, Crop Disease Detection, Hybrid Quantum-Classical Networks, Convolutional Neural Networks, Precision Agriculture, Parameterized Quantum Circuits, Agricultural Computer Vision.**

I. Introduction

Crop diseases are a major challenge to food security worldwide. The loss from these diseases is estimated at 20-40% of the total agricultural production each year. Manual inspection is the traditional method of a disease detection. This method is very labor-intensive,and also it takes a lot of time and is often unreliable since human error and subjective judgment are involved [1]. The use of this computer vision and deep learning for automated solutions has been developing very well but these methods are still limited by their computational efficiency and generalization capabilities[2].

The most recent advances in quantum computing are likely to become a paradigm shift in the classical machine learning algorithms and the latter would be the ultimate benefactors of such technological advances. Quantum machine learning (QML) calculates the computations of quantum mechanics using the laws of superposition, entanglement, and interference to calculate computations in a completely new way compared to classical computers[5].

One of the main reasons for the interest of many researchers in quantum computing applications in different fields has been the envisagement of the future. However, the application of quantum-enhanced deep learning to agricultural problems is still very much unexplored[7][8]. The only existing quantum approaches in agriculture are the ones that have mainly concentrated on quantum annealing for optimization problems or the ones that rely on simple quantum classifiers for processed features. The adoption of quantum processing directly in the agricultural computer vision pipelines is a research gap that has not been covered yet and that also has great practical implications.

This paper presents a novel orthogonal polynomial-based quantum attention network (OPQAN) method specifically designed for crop disease detection. Our main contributions include: a novel architecture integrating classical feature extraction with quantum processing modules, empirical validation on the PlantDoc dataset.The given approach shows better results compared to classical baselines and studies of quantum improvement in agricultural tasks and realistic implications of applying quantum-aware solutions in precision agriculture[9].

The rest of this paper is arranged in the following way: Section 2 included the background research that was necessary to be familiar with the offered method. Section 3 provides comparisons of the works that are associated with the classical and the quantum computing as per the methodology of the plant disease detection. Section 4 displays the pipeline of the proposed OPQAN method. Section 5 provides the mathematical representation of the same AQALCE algorithm with feature fusion technique. The findings and comparative analysis are provided in section 6 and lastly is the conclusion and future direction of proposed research in section 7.

II. Background works

A. Orthogonal Polynomials Based Transform Coding

It is common practice to model a linear two-dimensional image formation system as a Cartesian coordinate–separable blurring process, in which the observed image I is created by superposing weighted point sources that correspond to the object function f. For image analysis, it is especially helpful to express the object function f in terms of spatial derivatives of the image I with respect to Cartesian coordinates. It is

assumed that the point-spread function $M(x, y)$ is a real-valued function defined over the domain $(x, y) \in X \times Y$, where X and Y are ordered subsets of real numbers. The set X, which represents image rows for a grayscale image of size $n \times n$, is finite and can be indexed as $0, 1, \ldots, n-1$. The function $M(x, y)$ can be represented as a sequence under this discretization.

$$M(i, t) = u_i(t), \quad i, t = 0, 1, \ldots, n-1 \tag{1}$$

The 2D transform is given by

$$\beta'(\zeta, \eta) = \int_X \int_Y M(\zeta, x) M(\eta, y) I(x, y)\, dx\, dy \tag{2}$$

where β' represents the transform coefficients.

In discrete form, this becomes

$$|\beta'_{ij}| = (|M| \otimes |M|)' |I| \tag{3}$$

where $|M|$ represents the *matrix of orthogonal basis functions*, and $\otimes$ denotes the *outer product*.

The matrix $|M|$ is constructed using orthogonal polynomials $u_0(t), u_1(t), \ldots, u_{n-1}(t)$, as shown:

$$|M| = \begin{bmatrix} u_0(t_1) & u_1(t_1) & \cdots & u_{n-1}(t_1) \\ u_0(t_2) & u_1(t_2) & \cdots & u_{n-1}(t_2) \\ \vdots & \vdots & \ddots & \vdots \\ u_0(t_n) & u_1(t_n) & \cdots & u_{n-1}(t_n) \end{bmatrix} \tag{4}$$

The orthogonal polynomials are generated recursively as

$$u_{i+1}(t) = (t-\mu) u_i(t) - b_i(n) u_{i-1}(t), u_1(t) = t-\mu, u_0(t) = 1 \tag{5}$$

where

$$b_i(n) = \frac{i^2(n^2 - i^2)}{4(4i^2 - 1)}, \qquad \mu = \frac{1}{n} \sum_{t=1}^{n} t = \frac{n+1}{2} \tag{6}$$

These recursive relations ensure the orthogonality of the polynomial set, enabling them to act as valid basis functions for image transform coding. Point-spread operators $|M|$ of varying sizes can be constructed from Eq. (4) using the associated orthogonal polynomials for $n \geq 2$ by setting $t_i = i$. For ease of implementation in point-spread operations, the elements of $|M|$ are scaled so that all entries are integers.

B. The orthogonal polynomial basis

To simplify the computational process, the finite Cartesian coordinate sets X and Y are indexed as $\{1, 2, 3, \ldots, n\}$. The point-spread operator introduced in Eq. (3), which defines a linear orthogonal transformation for image coding, can be constructed as $|M| \otimes |M|$. The operator $|M|$ is obtained and suitably scaled using Eqs. (4) and (5).

We now demonstrate that a collection of $n \times n$ polynomial operators $(n \geq 2)$ constitutes a valid basis by satisfying the properties of completeness and linear independence. To establish this, consider the following orthonormal transformations. Under this framework, the image can be represented through a linear transformation as follows:

$$|Z| = |H^T||I| = (|M^T||M|)^{-1/2} |M^T||I| = (|M^T||M|)^{-1/2} |\beta'| \tag{7}$$

Here, $|H|$ is a *unitary operator*, ensuring energy preservation:

$$|H| = (|M| \otimes |M|) \times ((|M| \otimes |M|)^T (|M| \otimes |M|))^{-1/2} \tag{8}$$

Thus, the image reconstruction becomes:

$$|I| = |H||Z| = |M||\beta| = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \beta_{ij} |O_{ij}| \tag{9}$$

Bessel's equality confirms that the orthogonal transform is complete:

$$\langle |I|, |I| \rangle = \langle |Z|, |Z| \rangle \quad \text{(Bessel's Equality)} \tag{10}$$

or equivalently,

$$\sum_{i=0}^{n-1} \sum_{j=0}^{n-1} I_{ij}^2 = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} Z_{ij}^2 \tag{11}$$

This ensures that the total energy in the image is conserved after transformation. Any image or function $I(t)$ can be approximated by a linear combination of orthogonal polynomials:

$$I(t) = \sum_{1 \leq i \leq p} \alpha_i u_{i-1}(t) \tag{12}$$

The coefficients are determined by the *least squares method*:

$$\alpha_i = \frac{\sum_{1 \leq t \leq n} I(t) u_i(t)}{\sum_{1 \leq t \leq n} u_i^2(t)} \tag{13}$$

The polynomial basis operators of size (2×2) are:

$$O_{00}^2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad O_{01}^2 = \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}, \quad O_{10}^2 = \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}$$

The Polynomial basis operators of (3×3) are:

$$O_{00}^3 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad O_{01}^3 = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix},$$

$$O_{10}^3 = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \quad O_{11}^3 = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix},$$

$$O_{20}^3 = \begin{bmatrix} 1 & 1 & 1 \\ -2 & -2 & -2 \\ 1 & 1 & 1 \end{bmatrix}, \quad O_{21}^3 = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 0 & -2 \\ -1 & 0 & 1 \end{bmatrix},$$

C. Quantum-Inspired Qubit Extraction Method

Quantum-inspired learning frameworks map classical medical image features into quantum state representations to exploit principles such as superposition, rotation-based encoding, and inter-feature correlation modeling in a high-dimensional Hilbert space. A qubit is represented as a linear combination of basis states as indicated in Eq(14).

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \text{where } |\alpha|^2 + |\beta|^2 = 1, \tag{14}$$

ensuring valid probability amplitudes.

In plant leaf disease analysis, enhanced disease-related features such as color, texture, shape, and deep representations are first normalized and then encoded into quantum-inspired states. Given a classical feature vector $\mathbf{x} = [x_1, x_2, \ldots, x_d]$, the corresponding Hilbert space encoding is defined as represented in Eq(15).

$$|\psi\rangle = \sum_{i=1}^{d} \tilde{x}_i |i\rangle, \quad \tilde{x}_i = \frac{x_i}{\|\mathbf{x}\|}. \tag{15}$$

This superposition-based representation enables simultaneous modeling of multiple plant image characteristics.

To parameterize the feature-to-qubit transformation, each enhanced feature is encoded using rotation-based quantum-inspired gates according to Eq(16).

$$|\psi_{ij}\rangle = R_z(\theta_{ij})\, R_y(\phi_{ij})\, |0\rangle, \tag{16}$$

where θ_{ij} and ϕ_{ij} are rotation angles derived from Plant-specific feature values. This encoding allows continuous Plant image features to be efficiently mapped into qubit-like states.

To model inter-feature dependencies, entanglement-inspired correlations are introduced through controlled rotation mechanisms. The correlation between two encoded feature states is extracted as outlined in Eq(17).

$$F_{ij} = \langle Z_i \otimes Z_j \rangle, \tag{17}$$

which is conceptually analogous to the application of an $RZZ(\theta)$ gate in quantum systems. This formulation enables correlated plant image patterns to be jointly represented.

Finally, similarity between two encoded plant image samples is computed using a quantum-inspired kernel defined as specified in Eq(18).

$$K(\psi_i, \psi_j) = |\langle \psi_i | \psi_j \rangle|^2, \tag{18}$$

which facilitates nonlinear separation in the induced Hilbert space and serves as the theoretical foundation for the proposed hybrid quantum convolution neural network model based plant disease classification.

III. Related Work

Bairwa et al. (2024) proposed an image-processing-based scheme on pre-processing, segmentation, and hand-crafted feature extraction for tomato plant disease detection as a lightweight computational solution that can easily be adopted in agriculture, but traditional features make it not robust under different conditions in real-world fields to achieve maximum accuracy without advanced deep-learning-based feature learning that is present within today's Convolutional Neural Networks (CNN) and transformer models[1]. Therefore, this system will generalize poorly with no severity estimation and practical scalability.

Kumar et al. (2025) proposed an artificial intelligence–driven deep learning framework for plant leaf disease detection, employing CNN-based models, including DenseNet121 and InceptionV3 architectures [3]. All these powerful feature extraction methodologies classify plant diseases from both the Coastal region and Malnad region of Karnataka state. The advanced convolutional models ensure accuracy as well as robustness under any diverse environmental condition. But this method wants to have high computational resources that are not available in low-power agricultural devices; besides, due to limited region-specific datasets, model performance varies hence generalizability is reduced in other geographical environments.

Joshi et al. (2021) highlighted the significant advancements achieved by deep learning techniques, particularly CNNs, in image-based agricultural applications involving plant analysis [4]. In their study, CNNs by being able to learn hierarchical visual features from the data itself, were able to outperform previous approaches in terms of accuracy and scalability in comparison to image-processing approaches. They noted some practical challenges for limiting deployment for these methods on real farms. Another challenge for real-world deployment on low-computation resources on farms is the challenge of high computational costs and lesser explainability.

Fan et al. (2024) presented a hybrid quantum-classical convolutional neural network architecture of image classification, bridging between classical CNN designs and the new quantum computing technologies to improve the computational efficiency and image classification performance[5]. In this approach, quantum computing concepts are leveraged to accelerate feature extraction, while classical CNNs are employed for the final classification stage. Their study gives hopes that in future, it may be possible to reduce the computational cost when dealing with a large number of images. However, quantum computing depends on hardware which can have issues related to accessibility as well as scalability. The practical implementation of QC-CNN models is also constrained by the very youthfulness of a quantum computing ecosystem.

Lawrence and Ogedebe (2024) present a CNN model for plant leaf disease image recognition[6]. The process of using CNNs demonstrates the ability to automate features extraction from leaf images something that is a huge advantage over traditional methods. This system may address issues regarding agricultural diagnostics through promoting sustainable farming practices. Nonetheless, the potential success of this model may be hindered through the need for large high-quality labelled datasets and computational resources. Generalizing the model across a variety of plant species and environmental growing conditions may pose as a potential limitation.

Cruz et al. (2025) proposed a quantum image classification

system to boost the sustainability of the diagnosis of leaf images through improved computational efficiency [10]. The system uses quantum computing principles to significantly increase the speed of feature extraction and classification compared to traditional models and shows promise of using quantum algorithms to decrease processing time while maintaining diagnostic accuracy. However, the approach is limited by the availability of capable quantum hardware, in addition to the difficulty to implement hybrid quantum systems. Lastly, the implementation scalability and sustainability in real-world applications will continue to be a challenge as quantum computing is still an emerging set of technologies.

Wang et al. (2012) utilized neural networks for the classification of plant diseases using images, referring to AI methods within agronomy and agricultural diagnostics [13][14]. The authors created an algorithm that automated the feature extraction of leaves from an image, which was more efficient than manually examining the leaf. In general, Wang show that neural networks can accurately classify some of the different plant diseases. Limits include small datasets from their study and a relatively simple architecture of the networks. When accounting for other field conditions such as lighting and background, the performance may decrease even more.

Rose Anandhi and Sathiamoorthy (2023) proposed a method based on deep learning for classification and recognition of rice plant disease, using CNN architectures to improve diagnostic accuracy[15]. A holistic and automated approach was taken by the model for feature extraction and classification purposes to reduce to wholly removing human inspection. The model demonstrated accuracy on classification work on rice leaf images, showing the potential for the AI-precision agriculture model in properly diagnosing disease in a rice plant. A limitation of this method is the need for a large well-labeled dataset of images, and significant computational power for model training. Generalizability may also be an issue; environmental and rice type differences may reduce overall applicability.

Aiswarya K and Sreekumar N R (2022) proposed a quantum image processing approach for plant disease detection, combining quantum computing principles with image analysis to enhance classification efficiency [24]. Their method aims to improve feature extraction and processing speed compared to classical image-processing techniques. The study opens the potential use of quantum algorithms in agricultural diagnostics, particularly on large image datasets. However, practically it cannot be implemented due to the fact that quantum hardware is not available and also because of the complexity of the systems. Moreover, scalability and real-world deployment are issues because quantum computing technology is at an early stage.

IV. The Proposed OPQAN Method

A. Research Design

The proposed study was conducted from May 2009 to April 2014 in two phases. The first phase of this study focused on general plant disease image analysis, classification performance, and responses of agricultural researchers and plant pathology experts from various agricultural universities and research institutes in Chennai, Tamil Nadu, India. In this research, a new hybrid classical–quantum prototype model was developed to support plant leaf disease classification. In the second phase of this study, the experts' reactions to the overall concept of the proposed hybrid classical–quantum framework were obtained. Based on the experts' feedback, the initial prototype architecture, feature fusion strategy, and implementation approach were refined and enhanced using updated computational techniques and quantum-inspired methodologies.

B. The OPQAN Framework

The proposed OPQAN method consists of multiple layers including classical feature extraction, orthogonal feature space fusion, quantum feature encoding, and a quantum-inspired classification engine. Our method automatically extracted statistical, texture, and deep learning features from leaf images to support diagnosis healthy or diseased. The Fig. 1 shows the input and output of the proposed OPQAN framework.

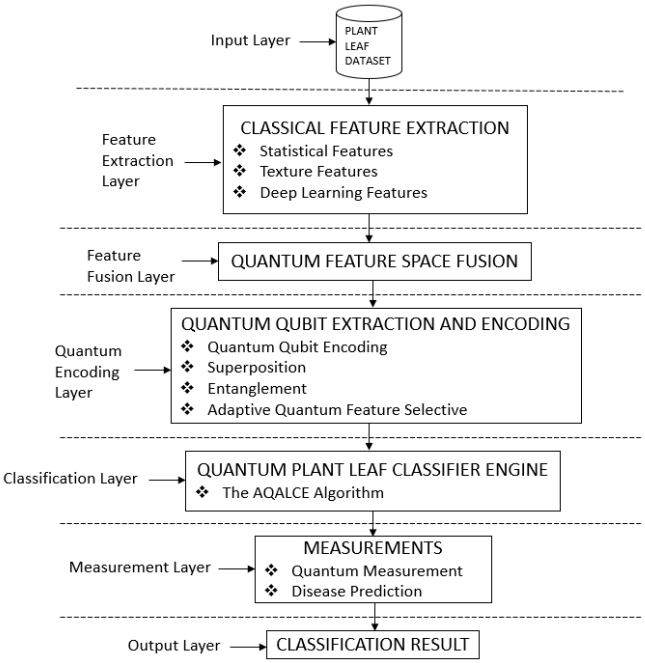

Fig. 1. The pipeline of the proposed OPQAN framework.

Algorithm 1 The Proposed OPQAN Algorithm

Input: Plant leaf image Dataset D
Output: Classification result (Healthy / Diseased)

1: Extract classical features from training image dataset D
2: Execute OFSF method
3: Apply quantum feature extraction method
4: Execute AQALCE Algorithm
5: Return diagnosis keyword

Fig 1 shows the input and output of the proposed Algorithm 1. In Step 1, classical feature extraction is performed on the

dataset D, where discriminative features such as statistical, texture, and deep learning techniques are extracted to capture the visual properties of the leaves. In Step 2, the OFSF method is applied to the extracted features to eliminate redundant and irrelevant information while retaining the most significant attributes, thereby improving computational efficiency and accuracy. In Step 3, a quantum feature extraction method is employed to transform the optimized features into a higher-dimensional quantum-inspired feature space, enhancing feature representation and improving class separability.

In Step 4, the AQALCE (Associative Quantum Attention-Based Leaf Classifier Engine) algorithm is executed, where the optimized and quantum-enhanced features are used to train the classification models and generate predictions. The classifier learns the distinguishing patterns between healthy and diseased samples and performs the final classification. Finally, in Step 5, the algorithm returns the diagnosis result, outputting whether the given plant leaf image is Healthy or Diseased.

V. The Mathematical Model of the Proposed OPQAN Algorithm

The proposed orthogonal polynomial-based quantum attention network (OPQAN) framework is designed to improve plant leaf disease classification by effectively integrating classical feature diversity with quantum-assisted learning. As illustrated in Fig. 1, the framework consists of a five major stages: classical feature extraction, orthogonal feature space fusion, quantum qubit encoding, quantum attention-based classification, and measurement-based decision making[16][17].

Initially, the input plant leaf image I undergoes classical feature extraction to capture complementary visual characteristics. Statistical features $\mathbf{F}_s$ represent pixel-level intensity distributions, texture features $\mathbf{F}_t$ describe spatial relationships among neighboring pixels based on orthogonal polynomial model, and deep learning features $\mathbf{F}_d$ are extracted using a pretrained convolutional neural network to encode high-level semantic information. These heterogeneous features are concatenated to form a unified feature vector as shown in Eq(19).

$$\mathbf{F}_c = [\mathbf{F}_s, \mathbf{F}_t, \mathbf{F}_d] \tag{19}$$

The concatenated feature vector is normalized to the range $[0, 1]$ to ensure numerical stability during subsequent processing.

To reduce redundancy and enhance discriminative capability, Orthogonal Feature Space Fusion (OFSF) is applied. The normalized feature vector is mapped onto orthogonal K-order polynomial basis. Calculations of the orthogonal coefficients are as shown in Eq(20).

$$\alpha_k = \sum_{i=1}^{N} F_c(i)\,\phi_k(i), \quad k = 0, 1, \ldots, K \tag{20}$$

where $\phi_k(\cdot)$ denotes the k-th orthogonal polynomial basis function. The orthogonality condition is defined as defined in Eq(21).

$$\sum_{i=1}^{N} \phi_k(i)\phi_l(i) = \begin{cases} 1, & k = l \\ 0, & k \neq l \end{cases} \tag{21}$$

The energy of each orthogonal component is calculated as illustrated in Eq(22).

$$E_k = \alpha_k^2 \tag{22}$$

Those dominant components whose energies are bigger are retained to constitute the orthogonal fused feature vector OFSF hence reducing redundancy and enhancing class separability. The chosen orthogonal features are then coded. Quantum qubit encoding of angle-based quantum qubit encoding into quantum states. Each of the chosen features $\mathbf{F}_{\text{OFSF}}$ is associated with its qubit. And the state is represented as formulated in Eq(23).

$$|\psi_i\rangle = \cos(F_{\text{OFSF}}^{(i)})|0\rangle + \sin(F_{\text{OFSF}}^{(i)})|1\rangle \tag{23}$$

The overall quantum state is expressed as shown in Eq(24).

$$|\psi\rangle = \sum_{i=0}^{2^n-1} \beta_i |i\rangle \tag{24}$$

In which n is the number of qubits and β_i are the encoded qubit amplitudes. Hadamard gates are used to introduce the concept of superposition and controlled quantum gates are used to establish entanglement to capture inter-feature dependencies. The formalized quantum feature selection An adaptive quantum feature selection mechanism additionally narrows down the encoded quantum features by silencing less informative quantum states.

It is classified by the Quantum Attention-Based Leaf Classifier (QALC), which uses a parameterized variational quantum circuit, in which an attention mechanism is applied to give importance weights to quantum features and is defined as presented in Eq(25).

$$\mathbf{A} = \text{Softmax}(W \cdot \mathbf{Q}) \tag{25}$$

Q is a quantum feature and W is trainable parameters. The measured quantum state is passed through a variational quantum circuit which can be expressed as indicated in Eq(26).

$$|\psi_{\text{out}}\rangle = U(\boldsymbol{\theta})|\psi\rangle \tag{26}$$

In which $U(\boldsymbol{\theta})$ is a unitary operator that is learned as part of the training.

Lastly, quantum measurement operations are carried out on the output quantum state. The likelihood of the input sample falling in class c is determined based on Eq(27).

$$P(c) = \langle\psi_{\text{out}}|M_c|\psi_{\text{out}}\rangle \tag{27}$$

where M_c is the operator of measurement of the class c. The last decision in classification is determined as expressed in Eq(28).

$$\hat{y} = \arg\max_c P(c) \tag{28}$$

VI. EXPERIMENTATION AND RESULTS

A. Dataset Description

The PlantDoc Dataset is a real-world plant leaf disease dataset designed for visual plant disease detection and classification. About 2,598 pictures of leaves taken from the internet sources like Google Images and Ecosia have been used. There are complex backgrounds and different lighting conditions in the field, not in a laboratory. Thirteen different plant species are presented along with seventeen disease categories showing a very big variety of symptom manifestations across apple, corn, tomato, potato, and others. The dataset was manually curated and annotated through. cannot human effort for labeling accuracy, and it offers a meaningful benchmark to train and assess computer vision as well as machine learning models towards identifying plant disease. The diversity of the dataset together with its real-world image complexity makes it very attractive for developing models that generalize well outside of optimal laboratory conditions.

B. Performance Measures

Standard performance measures are used to rate how well the model can tell healthy and diseased leaf samples apart in many plant species. As there are several disease classes with different sample shares, the rating must cover both between-class swings and class imbalance. Metrics which includes accuracy, precision, recall, and F1-score are embraced to quantify predictive capability. Accu- Accuracy provides an overall estimate of how well samples were classified, while precision measures for a particular class how true the positive predictions will be. Recall measures the ability to find all possible positive instances; F1-score gives the harmonic mean of both precision and recall which is more informative on imbalanced data than plain accuracy because it takes into the account of both false positives and false negatives.

In order to determine the performance of disease classification models quantitatively, the standard measures of performance are used. Such measures evaluate the ability of the classifier to distinguish the classes of plant diseases and the non-disease healthy classes. Where TP, TN,FP and FN is the number of True Positives, True Negatives, False Positives and False Negatives respectively as obtained in Eq(29)-(32).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{29}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{30}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{31}$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{32}$$

In multi-class plant disease classification problems, these metrics are generally computed on a per-class basis and then averaged across all classes to summarize the classifier's overall.The analysis of a confusion matrix helps to see how well predictions match up with different disease labels and to determine patterns of misclassification between similar symptoms of the diseases and the vegetation species. This especially applies to the case of the PlantDoc dataset, because it consists of complicated real-life imagery, with shifts in background, light, and symptom location. By using these performance measures, it is possible to objectively compare the various classical and quantum-based feature extraction, fusion, and classification methods put forward in the literature to recognize plant leaf disease with automated methods.

TABLE I
PERFORMANCE COMPARISON ON THE PLANTDOC DATASET

Method	Acc.	Prec.	Rec.	F1
QSVM	92.48	91.96	91.42	91.69
QNN	93.85	93.21	92.87	93.04
VQC	94.62	94.10	93.78	93.94
Hybrid CNN–QSVM	95.31	94.86	94.22	94.54
OPQAN	**98.17**	**97.92**	**97.65**	**97.78**

C. Experiment on PlantDoc Dataset

The PlantDoc dataset selected was experimented to determine the efficiency of the suggested OPQAN structure of plant leaf disease classification. The data was processed by downsizing the pictures, normalizing the color intensities, and pre-treating the samples to extract features so that they turned out to be good in training and testing. The dataset has a variety of species of plants and disease categories; hence, it is versatile enough to evaluate multi-class classification. The information was split into training and testing data to enable the model to learn disease patterns throughout training and determine how it fared in generalization throughout testing.

During experimentation, statistical, texture, and deep learning features were extracted from leaf images using statistical, OPT and CNN methods. These features were then fused and quantum encoded to enhance representation before classification. The model was trained to detect subtle and complex disease patterns present in real-world plant images. The dataset included multiple plant species and disease categories to ensure comprehensive evaluation. After training, the model was tested to evaluate its generalization capability. Performance was measured using standard evaluation metrics such as accuracy, precision, recall, and F1-score.

Table I demonstrates a consistent and progressive improvement in performance from standalone quantum models to hybrid and attention-based architectures.

D. Result Discussion

As shown by the performance analysis in Table I, the proposed model of the overall structured and functioning of the suggested system, the OPQAN, provides better diagnostic the capabilities when using the plant dataset (PlantDoc) compared to the current quantum-based classifiers. Although more basic algorithms, including QSVM and QNN, delivered decent results, their recognition rates were also influenced by the differences in the textures of the leaves, the nature of a symptom, and the complexity of the background of

the dataset. The VQC and Hybrid CNN-QSVM did better because of the improved feature learning and hybrid inference processes, which have shown that there is the advantage of using deep feature representations in quantum decision frameworks.Nevertheless, the suggested OPQAN model has always performed better against the reference approaches in all the performance indicators.

The orthogonal fusion of feature space led to more discriminative feature embeddings whereas the quantum attention-based leaf classifier enabled successful separation in disease categories. This made it possible to introduce inter-class similarities to the model as well as subtle visual differences in the symptom presentation of leaf diseases. All in all, the findings prove that the developed technique is more applicable to the real-life plant disease classification challenges and offers greater robustness, better generalization, and increased predictability compared to the currently available quantum-based methodologies.

TABLE II
ABLATION ANALYSIS OF THE PROPOSED OFSF-QALC FRAMEWORK

Configuration	**Accuracy (%)**
Without OFSF	95.84
Without Quantum Attention	96.21
Without AQFS	96.48
Full OFSF-QALC	**98.17**

To further evaluate the contribution of individual components within the proposed OPQAN framework, an ablation study was performed. This analysis involved systematically excluding selected modules and assessing the resulting impact on classification accuracy. From Table II, when the OFSF module is removed, the accuracy decreases to 95.84%, highlighting its importance in reducing feature redundancy and improving discriminative representation before quantum encoding. When the Quantum Attention mechanism is excluded, the accuracy drops to 96.21%, indicating its role in prioritizing disease-relevant qubit interactions. Similarly, removing the attention-based leaf classifier (AQALCE) results in an accuracy of 96.48%, demonstrating its contribution to eliminating irrelevant quantum features and enhancing robustness.

The highest accuracy of 98.17% is achieved with the complete OPQAN framework, confirming that the synergistic integration of orthogonal fusion, quantum encoding, and attention mechanisms significantly improves plant leaf disease classification performance.

VII. CONCLUSION AND FUTURE ENHANCEMENT

This research introduced a new hybrid OPQAN system to support plant leaf disease classification. The suggested method successfully unites orthogonal feature space fusion with a Quantum Attention Network(QAN) based classification to resolve the main problems of feature redundancy, as well as, inefficiency and low feature prioritization of the currently existing methods.The OPQAN module improves how classical data is prepared for quantum processing. It converts different types of classical features (like statistical, texture, or deep learning features) into a unified, orthogonal format. Because the features become compact and non-redundant, fewer qubits are required for quantum encoding. Moreover, quantum attention mechanism selectively amplifies disease-sensitive qubits, which allows exposing more expressive decision boundaries in variational quantum circuits. The proposed OPQAN method tested with the plantdoc dataset and displays an excellent performance, which reaches 98.17% the classification accuracy. Our method outperforms the current quantum-based classifiers, which justifies the usefulness of the proposed design.

Although the proposed framework performs effectively, further improvements can enhance its capability. Future work may extend the OPQAN model to multi-spectral and hyper-spectral plant images to capture deeper physiological characteristics. Dynamic optimization of quantum circuits and hybrid classical–quantum attention mechanisms can be explored to reduce computational overhead and improve performance under varying conditions. Additionally, cross-dataset evaluations and explainable quantum learning techniques can enhance generalization, interpretability, and real-world applicability in agricultural decision support systems.

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